From Lie algebras to Frobenius's Theorem

Algebraic Formalisation with locales, types and relations

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Lie Groups and Algebras

- A Lie group is a manifold that is also a group under a smooth operation $(x, y) \mapsto xy$ with smooth inverses $x \mapsto x^{-1}$.
- A Lie algebra is anticommutative and obeys the Jacobi identity:

$$\forall x, y, z \in \mathfrak{g} : \qquad [x, [y, z]] + [y, [z, x]] + [z, [x, y]] = 0$$

locale
$\mathrm{lie}_{g}rp = c_{m}anifoldchartsinfinity > +grp_{o}ncarriertmsone + assumessmooth_{m}ult :$
$"diff_{o}n_{p}roduct_{m}anifoldchartstms"\ andsmooth_{i}nv : "diffinfinity > chartschartsinvs"$

Type Classes: Polymorphism in Isabelle/HOL

Quantification over types is not possible in HOL.

$< 1|handout : 0 > \alpha \exists a \in (\mathrm{UNIV} :: \alpha\ \mathrm{set}) < 2|handout : 0 > \alpha P(\alpha) \implies \exists a \in (\mathrm{UNIV}$

Type classes offer a (restricted) alternative. A class can be defined much like a locale: it is possible that no types satisfy the sort constraints. class $\mathrm{fin}_d im_r eal_v ector = real_v ector + fixesbasisassumesfinite_B asis :$ "$finitebasis$" $andindependent_B asis :$ "$nexists > u.(Sum > vin > basis.uv *^s ub > Rv) = 0and > (exists > vin > basis.uvnoteq > 0)$" $andspan_B asis :$ "$Sum > ain > basis.ra *^s ub > Ra|r.True = UNIV$"

Transfer: Relators

Relators allow us to build transfer rules for constants from transfer relations. The most ubiquitous one is the function relator.

$\alpha[d,"f"']['r,"\cong "]\beta[d,"g"]\ \alpha[r,"\cong "']\beta$

- α and β *related* by $\cong\colon \alpha \to \beta \to \texttt{bool}$
- Say a theorem involves a function $f\colon \alpha \to \alpha$, and we have a candidate $g\colon \beta \to \beta$ with equivalent behaviour
- Isabelle can use a transfer rule ¡4¿constructed from $(\cong)$ and the function relator:

$$<4> a, ba \cong b \implies f(a) \cong g(b)[(\cong) ===> (\cong)]\ f\ g$$

(Real) Division Algebras

A (associative) division algebra is an (associative) algebra that has multiplicative identity and inverses.
class $\mathrm{scalar}_a lgebra = real_d iv_a lgebra + fin_d im_r eal_v ector$

- is a division algebra over itself. $= \{a + bi + cj + dk \mid a, b, c, d \in\}$
- is a division algebra over . $i^2 = j^2 = k^2 = ijk = -1$
- is a division algebra over . not commutative: $ij = k$ but $ji = -k$

Polynomials over Type Embeddings

The fundamental theorem of algebra states that every complex polynomial has a root. This implies every complex polynomial can be factorised.

$$p_(x) = c \prod_{j=0}^{\deg(p)} \Big(x - r(j)\Big)$$

definition "c simeq¿ r equiv¿ c = (r *^{s} *ub* > *R*1)" *definition*" *pdoteq* > *qequiv* > (*forall* > *i* :: *nat*.*coeffpisimeq* > *coeffqi*)"

$$p_(x) = r \prod_{j=0}^{N_r} \Big(x - r_r(j)\Big) \prod_{k=0}^{N_i} \Big(x - r_i(k)\Big)\Big(x - \overline{r_i(k)}\Big)$$